

Spread Size, Outcome Variance, and the Illusion of Point Shaving in NCAA Basketball

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Abstract

This paper examines whether the divergence documented by Wolfers (2006) can identify point shaving in NCAA basketball. Wolfers uses the gap between a favored team's probability of winning without covering the point spread and its probability of winning by twice the spread as evidence of systematic manipulation in collegiate athletics. Using 95,781 Division I games spanning 2003 to 2025, I show that this identification strategy is fundamentally flawed, because the divergence between these two probabilities is a mechanical consequence of spread size rather than evidence of manipulation. Specifically, a team favored by S points has exactly $S - 1$ winning margins that fail to cover, so the probability of winning without covering rises mechanically with spread size even in the complete absence of point shaving. A simulation exercise confirms this formally: when margins of victory are drawn from a normal distribution centered on the spread with no manipulation imposed, the Wolfers divergence appears clearly in the simulated data.

Correcting for this asymmetric mechanical relationship with a heteroskedastic probit that explicitly models the increasing variance of game outcomes in spread size, I find no systematic evidence that larger favorites are more likely to win without covering, and estimating the model year by year across the two-decade sample reveals no time trend in the residual effect of spread size. As an extension, skew-probit models allow for asymmetry beyond what variance alone captures; these achieve superior fit on standard information criteria but are presented as exploratory, given the difficulty of interpreting the skewness parameter in a theoretically precise way. Taken together, the results suggest that existing tests for point shaving based on spread-coverage probabilities are misspecified, and that the evidence cited in prior work is more parsimoniously explained by the mathematical structure of point spread betting than by behavioral manipulation.

Keywords: point shaving; spread betting; market efficiency; heteroskedastic probit; skew-normal

1 Introduction and Background

In recent years the rise of sports gambling has been undeniable. For much of the twentieth century sports gambling was restricted within the United States, with the exception of Nevada. In 2018, however, the Supreme Court’s decision in *Murphy v. NCAA* struck down the long-standing ban, and the past decade has seen extreme growth within the sector as sports gambling has moved outside the casino world and into the lives of the average sports consumer.

With this rise comes a resurgence of interest in the efficiency of these gambling markets, and in how individuals may derive an edge either through betting strategy or through manipulation. With renewed interest in these topics comes renewed interest in the concept of point shaving. In this paper I reanalyze an approach established by Wolfers (2006) and argue that his primary identification strategy suffers from a fundamental mechanical flaw that renders his evidence of widespread corruption difficult to interpret.

Point shaving refers to the practice by which a player or players on a heavily favored team deliberately limit their winning margin to fall short of the point spread, allowing gamblers who have bet on the underdog to collect their winnings while the favored team still wins the game. The incentive structure driving this behavior is straightforward. In spread betting markets, gamblers do not bet on who wins the game outright but rather on whether the favored team wins by more than a specified margin, the spread. This creates a discontinuity in payoffs at precisely the point that should be largely irrelevant to the players themselves, since a player’s primary objective is winning the game rather than the margin of victory. Wolfers (2006) argues that this asymmetry between what players care about and what gamblers care about creates the conditions for a mutually beneficial corrupt contract, in which a gambler offers a payment to a player in exchange for the player reducing effort just enough to fall short of the spread while still winning the game.

2 Data

The primary dataset consists of Division I NCAA men’s basketball game outcomes and betting line data drawn from Todd Beck’s Prediction Tracker catalogue¹. This database provides individual game scores for all Division I matchups spanning the 2003–04 through 2024–25 seasons, yielding a total of 95,781 game observations. Critically, the database also includes 42 distinct computational spread predictions generated by a variety of quantitative forecasting models for each game. These model-generated predictions allow the construction of an aggregated market spread measure that is not subject to the same potential concerns that might arise from using a single bookmaker’s line. The primary spread variable, `lineavg`, is constructed as the simple average across all 42 available computational predictions for each game, providing a robust market-based forecast of the expected winning margin.

From these raw game outcomes and spread predictions I construct several variables. The realized point differential for each game is computed as the home team’s score minus the visiting team’s score, and the absolute spread is defined as the absolute value of `lineavg`, denoted `absline`. From these foundational variables I construct three dummy variables. The first, `fav_won`, equals one if the favored team’s realized margin of victory is strictly positive, indicating the favorite won the game outright

¹<https://www.thepredictiontracker.com>

regardless of the spread. The second, `fav_covered`, equals one if the favored team's margin of victory strictly exceeds `absline`, indicating the favorite won by more than the spread. The third, `beat_2x`, equals one if the favored team's margin of victory strictly exceeds twice `absline`.

Finally, I generate the two primary outcome variables of interest. The first, denoted Y , equals one in games where the favored team won but failed to cover the spread. These are the games that would be characterized as suspicious in terms of possible point shaving. The second, Y_{2x} , equals one in games where the favored team beat the spread but did not cover twice the spread.

Table 1 presents summary statistics by season for the full sample of 95,781 NCAA Division I basketball games spanning 2003 to 2025. The number of observations per season grew considerably over the sample period, rising from 2,844 games in 2003 to over 5,600 by the mid 2020s, reflecting the expansion of Division I basketball and improved data coverage. The average point spread remains stable throughout the sample, hovering between 7.3 and 8.2 points across all seasons, suggesting that the composition of games in terms of competitive balance has not changed meaningfully over time. The average realized point differential similarly shows little secular trend, ranging between 4.2 and 5.8 points across seasons. Most relevant to the analysis, the proportion of games in which the favored team won but failed to cover the spread shows a modest but notable decline over the sample period, falling from roughly 24–26 percent in the early 2000s to 21–23 percent in more recent seasons. Whether this decline reflects changing game dynamics, improved market pricing, or is simply noise around a stable mean is a central question the empirical analysis addresses.

3 Empirical Strategy: Replicating Wolfers (2006)

The empirical strategy proceeds in four stages. The first replicates the graphical approach of Wolfers (2006) using the sample of NCAA basketball games spanning 2003 to 2025, constructing the two probability series at the heart of his identification strategy and confirming that the divergence he documents is present in more recent data as well. The second demonstrates formally that this divergence is a mechanical consequence of spread size rather than evidence of manipulation. The third estimates a series of regression models that correct for this mechanical relationship and tests whether any residual effect of spread size on the probability of winning without covering remains. The fourth examines whether the corrected estimates display any time trend across the two-decade sample, directly addressing the question of whether patterns in the data are consistent with changing rates of manipulation over time.

I begin by generating kernel densities of winning margins relative to the spread (Figures 1 and 2). If the spread is the market-generated forecast, the realized difference from the spread is the forecast error. These errors should be uniformly distributed about the spread for games both with and without heavy favorites. For games without heavy favorites—games where no team has the ability to take advantage of point shaving, shown in Figure 1 for games with `lineavg` below twelve—the distribution follows a normal shape. For games with heavy favorites, however, the distribution is slightly right-skewed, which would indicate “too few” strong favorites beating the spread and the occurrence of possible point shaving.

Wolfers (2006) identifies that this conclusion is preliminary, and that more convincingly proving the existence of point shaving would require a fully structural model of how players act while shaving, since

Table 1. Summary statistics by season

Season	<i>N</i>	% won, not cover	Avg. spread	Avg. point diff.
2003	2,844	24.40	7.83	4.98
2004	3,007	25.04	7.63	5.39
2005	2,964	23.92	7.34	4.60
2006	3,340	23.56	7.72	5.74
2007	3,431	23.75	8.20	5.77
2008	3,242	22.76	7.98	4.74
2009	3,574	26.50	8.23	4.73
2010	3,631	24.04	8.19	5.62
2011	3,573	25.02	8.00	5.38
2012	3,833	24.84	7.87	5.03
2013	3,875	24.08	7.75	4.82
2014	3,939	24.68	7.71	4.73
2015	3,940	22.99	7.78	5.04
2016	4,012	23.13	7.72	4.83
2017	4,077	22.61	7.68	5.12
2018	5,564	23.60	7.86	5.01
2019	5,288	22.67	7.85	5.02
2020	3,879	20.86	7.38	4.17
2021	5,303	23.91	7.90	4.61
2022	5,401	21.79	7.39	4.97
2023	5,622	21.66	7.74	4.81
2024	5,782	22.50	7.91	5.36
2025	5,659	23.27	8.15	5.27

Notes: “% won, not cover” is the share of games in which the favored team won outright but failed to cover the spread. “Avg. spread” is the mean of *absline*, the absolute value of the 42-model average forecast. “Avg. point diff.” is the mean realized margin. Full sample: 95,781 Division I games.

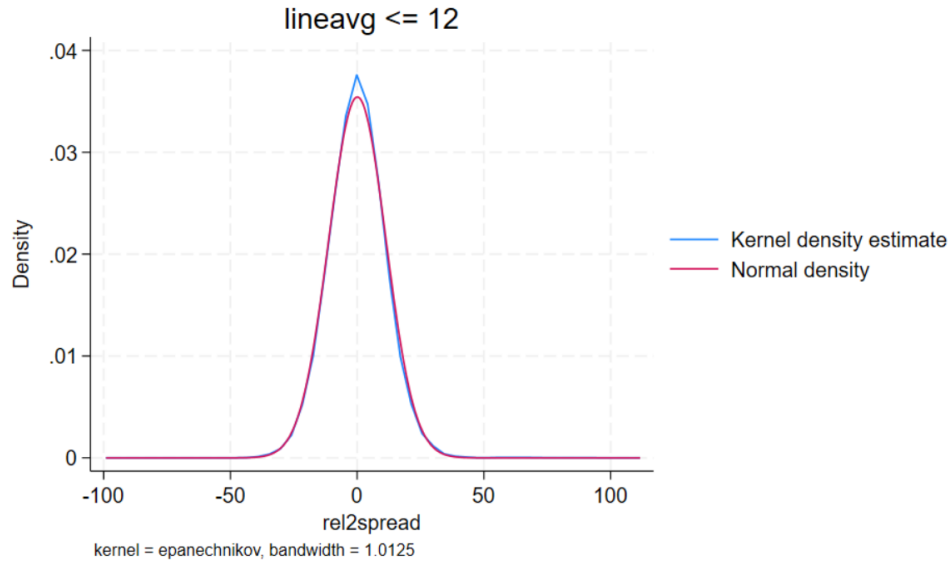


Figure 1. Kernel density of the forecast error—realized winning margin less the spread—for games with *lineavg* below twelve. The distribution is approximately normal, as a well-priced market implies.

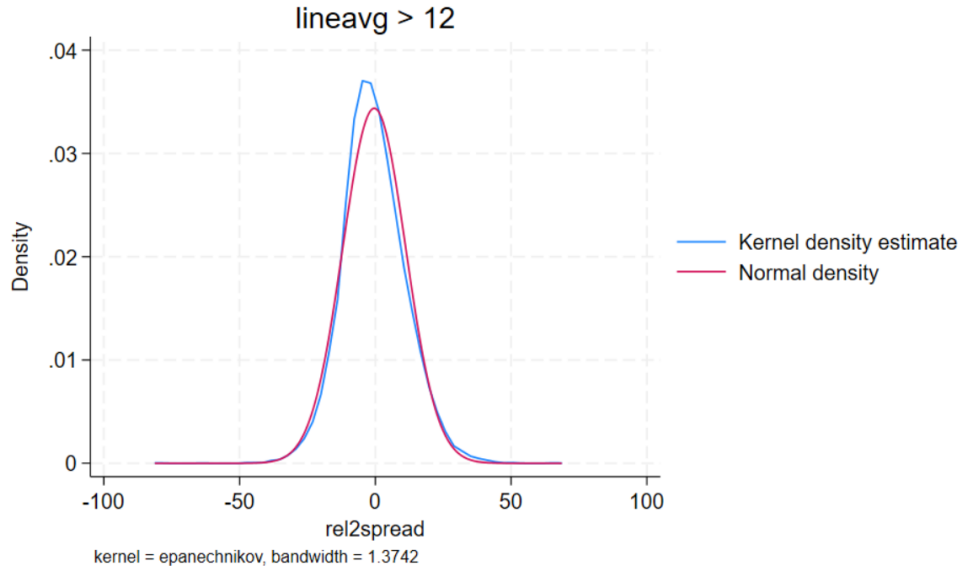


Figure 2. Kernel density of the forecast error for games with heavy favorites. The distribution is slightly right-skewed, the pattern read in prior work as “too few” heavy favorites covering.

the distribution of outcomes under point shaving cannot be observed. He follows this identification of the problem with a null-hypothesis scenario assuming the distribution of forecast errors is symmetric.

Figure 3 depicts these probabilities binned by spread bins of size two. As expected by construction, both probability series expand logarithmically as the spread S increases. This is expected structurally, since as S increases, the number of ways to win the game but not cover the spread, $S - 1$, increases. Importantly for the design of Wolfers (2006), the two probabilities diverge; under the null assumptions this would indicate a “suspicious” higher probability of heavily favored teams not covering the spread relative to the probability of covering twice the spread given that they beat the spread. This outcome would be consistent with a high prevalence of point shaving.

I build on this established work by exploring the prevalence of the divergence over time. Figure 4 depicts the Wolfers-style divergence within five periods: four five-year periods spanning 2003–2023, and one period after 2023. All the panels show the same general pattern of logarithmic growth, as well as the “suspiciously” lower probability of highly favored teams beating the spread relative to the probability of beating twice the spread given that they beat the spread. Crucially, however, the strength of this divergence has been shrinking over time, especially in the 2019–2023 series, where the lines reconverge around a spread of 23.

4 The Divergence Is Mechanical

Extending this null experiment, I formally demonstrate that the divergence documented in Figures 3 and 4 is a mechanical consequence of the structure of spread betting rather than evidence of behavioral manipulation. To do this I construct a simulation in which margins of victory are drawn from a normal distribution centered on the spread, with a standard deviation estimated from the residuals of a

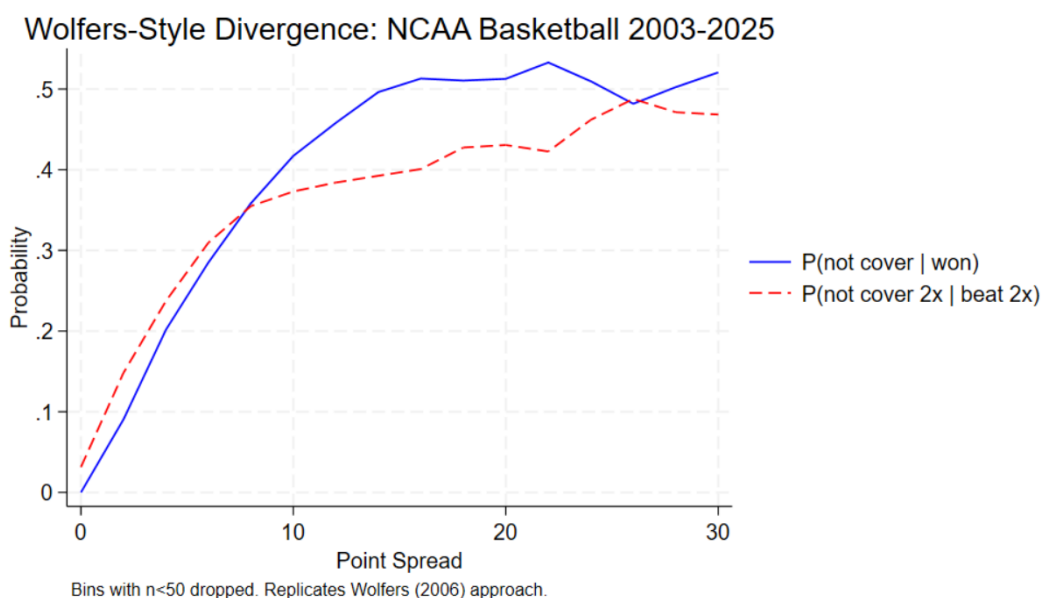


Figure 3. The Wolfers comparison replicated on 2003–2025 data: $P(\text{won but did not cover} \mid \text{won})$ against $P(\text{did not beat twice the spread} \mid \text{covered})$, binned by spread. Bins with $n < 50$ dropped.

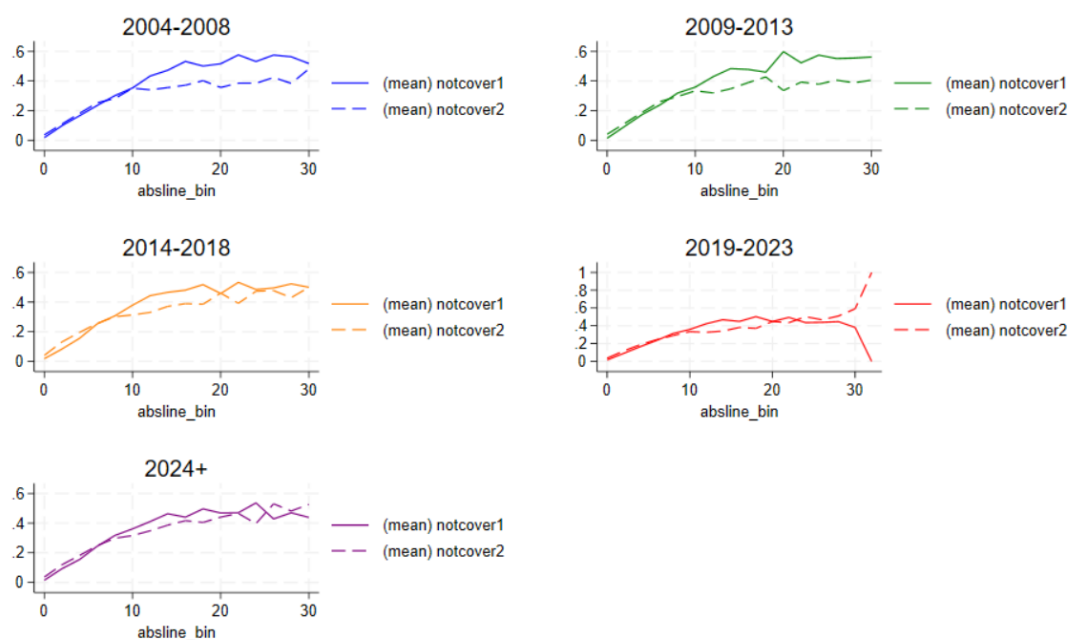


Figure 4. The Wolfers-style divergence estimated separately across five periods. The pattern repeats throughout the sample but narrows after 2019.

regression of realized winning margins on the point spread. This estimated standard deviation captures the typical forecast error in the sample after accounting for the linear relationship between spread size and realized margin. By construction this simulation contains zero point shaving, zero manipulation, and zero behavioral anomaly of any kind. Every outcome is generated purely from a symmetric normal distribution centered on the spread, meaning any pattern that emerges in the simulated data cannot possibly reflect corruption.

I simulate 50 replications of the full sample, drawing a margin of victory for each game from $N(\text{spread}, \hat{\sigma})$ and then applying the exact comparison of Wolfers (2006), computing $P(\text{won but not cover} \mid \text{won})$ and $P(\text{not beat } 2x \mid \text{beat spread})$ across spread bins. Figure 5 plots the average of these two probability series across all 50 replications.

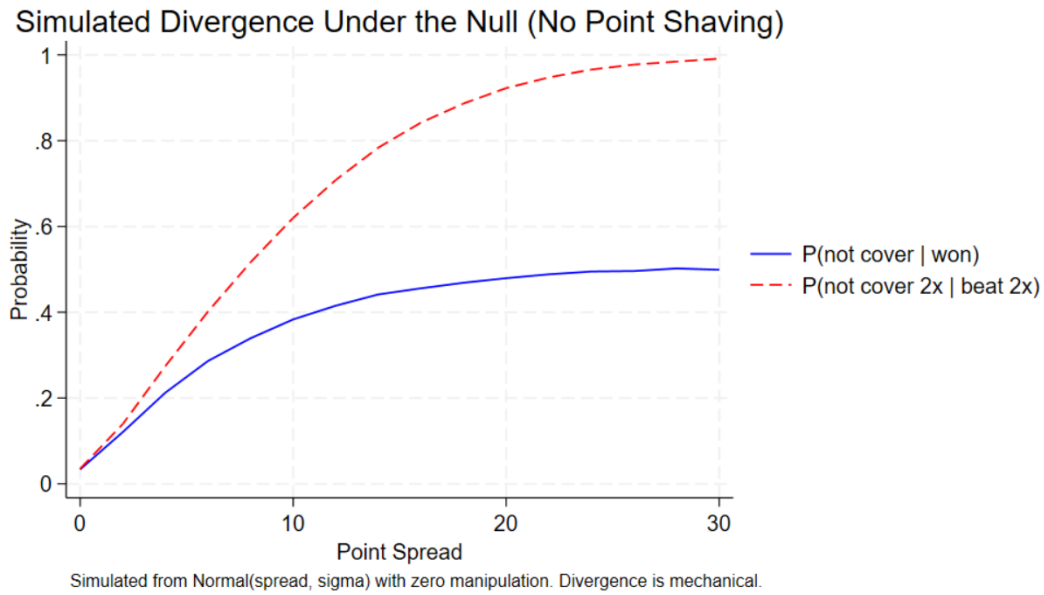


Figure 5. The central result. Fifty simulated replications of the full sample, with margins drawn from a normal distribution centered on the spread and zero manipulation imposed by construction. The divergence appears regardless.

Despite the complete absence of manipulation or point shaving by construction, a prominent divergence between the probabilities appears in the simulations, similar to what is observed in the real data. The reason is intuitive. As the spread S increases, the number of winning margins that constitute a win without a cover grows as $S - 1$, mechanically pushing the first series upward. The second series rises even faster, because beating twice the spread becomes an increasingly extreme threshold: at a spread of 20 a team must win by 40 points to beat twice the spread, which almost never occurs, meaning nearly every team that covers the spread also fails to beat twice the spread. The divergence between the two series therefore reflects nothing more than the arithmetic of point margins and the shape of the outcome distribution, and would appear in any dataset regardless of whether point shaving is occurring. Comparing the real data to the simulated null, if any conclusion were to be drawn it would be the opposite of the Wolfers-style reading—that favored teams too often cover the spread given that they win, relative to the probability that they beat double the spread given that they beat the spread.

I therefore reject the hypothesis that the Wolfers-style divergence of $P(\text{cover} \mid \text{win})$ and $P(\text{cover } 2x \mid \text{cover})$ is informative about the presence of point shaving.

5 Correcting for Variance and Asymmetry

Having established that the Wolfers divergence is mechanical and cannot be interpreted as evidence of point shaving, the natural question becomes whether any rigorous statistical approach can recover a meaningful signal from the data once the mechanical relationship between spread size and cover probability is properly accounted for. Testing this requires a model that explicitly separates the mechanical component from any potential behavioral component.

5.1 *Standard probit benchmark*

I begin the regression analysis by estimating a standard probit model as a benchmark, regressing the outcome variable Y_i , equal to one when the favored team wins but fails to cover the spread, on the standardized absolute spread and its square, conditional on the favored team winning. The quadratic term allows for a nonlinear relationship between spread size and cover probability. This specification takes the form

$$\Pr(Y_i = 1 \mid z_i) = \Phi(\beta_0 + \beta_1 z_i + \beta_2 z_i^2), \quad (1)$$

where $\Phi(\cdot)$ denotes the standard normal cumulative distribution function and $z_i \equiv z_i^{\text{absline}}$ is the average forecasted spread standardized to have mean zero and unit variance.

The standard probit serves as a useful baseline for two reasons. First, it confirms that spread size has a statistically significant raw relationship with Y in the data, consistent with both the finding of Wolfers (2006) and the mechanical relationship documented above. Second, it provides a log-likelihood benchmark against which more flexible specifications can be compared, allowing an assessment of how much variation in Y is left unexplained by a model that does not account for the mechanical properties of the spread distribution.

5.2 *Heteroskedastic probit*

The standard probit suffers from a fundamental limitation in this setting: it assumes that the variance of the latent outcome is constant across all spread sizes. Both the graphical evidence and the simulation above show this assumption to be implausible. Large-spread games generate fundamentally more variable outcomes due to garbage-time scoring, pulled starters, and reduced competitive intensity in the late stages of lopsided games. Imposing constant variance in this setting conflates the mechanical increase in outcome variance at large spreads with a genuine behavioral signal, which is precisely the misspecification that leads Wolfers (2006) to interpret the divergence as evidence of corruption.

To address this I estimate a heteroskedastic probit that relaxes the constant-variance assumption by modeling the standard deviation of the latent outcome as a function of spread size:

$$\Pr(Y_i = 1 \mid z_i) = \Phi\left(\frac{\beta_0 + \beta_1 z_i + \beta_2 z_i^2}{\exp(\gamma_1 z_i + \gamma_2 z_i^2)}\right). \quad (2)$$

The numerator, the mean equation, captures any residual level effect of spread size on the probability of not covering after accounting for variance. If Wolfers (2006) is correct and point shaving drives the divergence, β_1 should remain positive and statistically significant even after controlling for variance in the denominator. The denominator, the variance equation, explicitly models how the dispersion of outcomes changes with spread size, absorbing the mechanical relationship that invalidates the original identification strategy. The exponential functional form ensures the estimated variance is always positive, and the quadratic term in z_i allows for a flexible nonlinear variance profile across the spread distribution.

5.3 Skew probit

Finally, I explore a skew-probit framework as an even more flexible form. While the heteroskedastic probit provides a clean and theoretically motivated correction for the mechanical relationship between spread size and cover probability, it is restricted in one important respect: it assumes that the outcome distribution is symmetric around its mean after accounting for variance. That is, conditional on spread size, the heteroskedastic probit treats the probability of winning by more than expected as equal to the probability of winning by less than expected.

This symmetry assumption may itself be too restrictive. The late-game dynamics described above—garbage-time scoring, pulled starters, reduced defensive intensity—do not necessarily add variance to outcomes symmetrically. They may systematically push final margins in one direction relative to the spread, generating genuine asymmetry in the outcome distribution that the heteroskedastic probit cannot capture.

To explore this possibility I estimate a skew-probit model following the specification of Azzalini (1985), which augments the standard probit by introducing a skewness parameter α that allows the probability curve to be asymmetric. Letting the latent error follow a standard skew-normal distribution, the model takes the form

$$\Pr(Y_i = 1 | z_i) = \Phi(\eta_i) - 2T(\eta_i, \alpha_i), \quad \eta_i = \beta_0 + \beta_1 z_i + \beta_2 z_i^2, \quad (3)$$

where $\Phi(\cdot)$ is again the standard normal cumulative distribution function, η_i is the linear index capturing the effect of spread size on the probability of not covering, $T(\cdot, \cdot)$ is Owen's T function, and α_i is the skewness parameter. The model nests the standard probit as a special case: when $\alpha = 0$ the second term vanishes for all values of η , and the expression reduces to the familiar standard probit probability in equation (1). When α departs from zero the probability curve becomes asymmetric, rising steeply on one side and gradually on the other.

I proceed in two stages. In the first, I estimate a restricted version in which α is a single constant estimated from the data, with spread size and its square entering only the mean equation. This specification asks a simple but important question: is there evidence of systematic asymmetry in the outcome distribution on average across all spread sizes, after controlling for the level effect of spread size on the probability of not covering? A statistically significant constant α would indicate that the shape of the outcome distribution departs from symmetry in a way that neither the standard probit nor the heteroskedastic probit can capture, providing additional justification for the skew-probit framework on purely empirical grounds.

In the second stage I extend the model to allow α to vary as a function of spread size,

$$\alpha_i = \alpha_0 + \alpha_1 z_i + \alpha_2 z_i^2 + \alpha_3 z_i^3, \quad (4)$$

estimating a fully flexible skew probit in which both the level and the shape of asymmetry in the outcome distribution change across the spread distribution. This allows the degree and direction of skewness to shift as spread size increases, capturing the possibility that the nature of outcome asymmetry at moderate spreads, where games remain competitive and margins cluster tightly, differs fundamentally from outcome asymmetry at large spreads, where blowout dynamics, garbage time, and strategic substitution patterns dominate the final margin.

6 Results

Table 2 presents results from the four specifications, estimated on the sample of 70,207 games in which the favored team won. I discuss each specification in turn before drawing broader conclusions about the Wolfers hypothesis.

The standard probit in column (1) does find statistical significance in z_i , consistent with the finding of Wolfers (2006) that larger favorites are more likely to win without covering the spread. The coefficient of 0.668 with a t statistic of 87.09 would, under the Wolfers interpretation, be strong evidence of point shaving increasing with spread size. As the simulation exercise demonstrates, however, this significance is expected to arise mechanically even in the complete absence of manipulation.

The heteroskedastic probit in column (2) provides the clearest evidence for the central argument. Allowing the variance of the latent outcome to change with spread size, the coefficient on z_i in the mean equation drops from 0.668 to 0.197—a reduction of over 70 percent. Meanwhile the variance equation reveals that spread size is a powerful predictor of outcome dispersion, with a coefficient of 1.009 and a t statistic of 23.45. A substantial portion of what appeared to be a “point shaving signal” in the standard probit was in fact heteroskedasticity: larger-spread games produce more variable outcomes due to garbage-time scoring, strategic substitution patterns, and reduced competitive intensity in lopsided matchups. Once this mechanical source of variation is properly absorbed into the variance equation, the residual effect of spread size on the probability of not covering is dramatically attenuated.

Turning to the exploratory skew-probit specifications, allowing for asymmetry in the outcome distribution yields meaningful improvements in model fit. The constant- α skew probit in column (3) estimates a skewness parameter of -0.728 with a t statistic of -25.99 , indicating a statistically significant departure from symmetry in the distribution of outcomes conditional on spread size. The fully flexible skew probit in column (4), which allows the skewness parameter to vary as a polynomial function of spread size, achieves the best log-likelihood of all four specifications at $-38,962.4$, an improvement of 476 log-likelihood points over the standard probit. All coefficients in the α equation are highly significant, indicating that the degree of distributional asymmetry varies meaningfully across the spread distribution. This is the most notable methodological contribution of the paper: skew-normal distributions generate a better log-likelihood than standard probit specifications, highlighting the fundamental flaw in assuming symmetry when modeling spread outcomes. The standard probit, and even the heteroskedastic probit, impose a symmetry restriction that the data clearly reject.

Table 2. Model comparison: probit, heteroskedastic probit, and skew probit

	(1) Probit	(2) Het. probit	(3) Skew probit (const. α)	(4) Skew probit (varying α)
<i>Mean equation</i>				
z_absline	0.668*** (87.09)	0.197*** (9.11)	0.927*** (60.72)	0.379*** (23.08)
z_absline2	-0.167*** (-50.35)		-0.194*** (-35.86)	-0.0771*** (-14.85)
_cons	-0.467*** (-80.69)	-0.330*** (-47.68)	-0.627*** (-42.17)	-0.928*** (-95.94)
<i>Variance equation ($\ln \sigma$)</i>				
z_absline		1.009*** (23.45)		
z_absline2		-0.119*** (-5.94)		
<i>Skewness equation (α)</i>				
z_absline				-1.200*** (-15.75)
z_absline2				1.170*** (13.47)
z_absline3				-0.237*** (-7.69)
_cons			-0.728*** (-25.99)	-2.239*** (-16.40)
Log-likelihood	-39,438.4	-39,013.4	-39,166.7	-38,962.4
Observations	70,207	70,207	70,207	70,207

Notes: Dependent variable equals one if the favorite won but did not cover the spread. Sample restricted to games the favorite won. t statistics in parentheses.
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

7 Conclusion

Taken together, these results support several conclusions. First, the identification strategy of Wolfers (2006) is fundamentally flawed. The divergence between the probability of winning without covering and the probability of not beating twice the spread given covering is a mechanical consequence of how point margins relate to spread size, not evidence of behavioral manipulation. The simulation exercise confirms this by producing an identical divergence pattern from purely symmetric, manipulation-free data.

Second, there does appear to be genuine asymmetry in the distribution of spread outcomes, as evidenced by the significant improvement in fit from the skew-probit specifications. Similar to what Wolfers documented, some factor in the data produces left-skewness relative to a normal distribution. The original methodology used to attribute this pattern to point shaving was inherently flawed, however, and the source of the asymmetry is far from obvious. It could reflect the tendency of winning teams to reduce effort once a comfortable lead is established rather than running up the score. It could reflect improvements in betting-line accuracy over time as sportsbooks have adopted more sophisticated pricing models. It could reflect the strategic incentives facing coaches who rest starters in blowouts. Or it could reflect some combination of these and other factors that have nothing to do with corruption.

What can be said with confidence is that identifying point shaving from aggregate spread-coverage data requires far more careful econometric treatment than existing work has provided. The mechanical relationship between spread size and coverage probability must be explicitly modeled, the heteroskedastic structure of outcomes must be accounted for, and the symmetry of the outcome distribution cannot be taken for granted. Without these corrections, apparent evidence of point shaving is more parsimoniously explained by the mathematical structure of point spread betting itself.

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